

LESSON 7-4 • SESSION 1 • PRACTICE • FORM A

Comparing Numbers on a Number Line

Learning target – I can compare two rational numbers using their positions on a number line. • 6.NS.C.7a

REMEMBER

- On a number line, the number **farther right is greater**. Any positive number is greater than any negative number.
- For negatives, the one **closer to 0** is greater. Bigger digits do not mean a bigger negative number: $-13 > -40$.
- For decimals, use the same number of digits: $-0.06 > -0.60$.

WORKED EXAMPLE

Compare -14 degrees and -5 degrees.**Step 1** Plot -14 and -5 on a number line.**Step 2** -5 is to the right of -14 .**Step 3** Write the comparison: $-14 < -5$.**Answer** -5 is greater; -14 is colder**WORDS** **greater than** (*mayor que*) • **less than** (*menor que*) • **inequality** (*desigualdad*)

PRACTICE

show your work • label units

1. Write
- $<$
- or
- $>$
- in each blank.

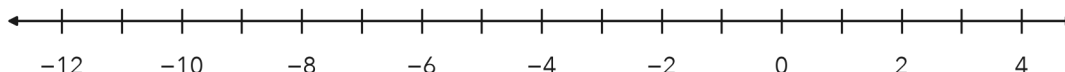
a. -4 ____ -9

b. -2.5 ____ 1.5

c. -6 ____ -2

d. 0 ____ -4

- 2.
- Trivia Scores.**
- Scores: Rosa
- -6
- , Tomas
- 3
- , Willa
- -11
- , Pia
- 0
- .

Hint The score farthest right is greatest.

- a. Plot and label all four scores.
- b. Compare Rosa and Willa using $>$. _____
- c. Greatest score ____ Least score ____

- 3.
- Two Accounts.**
- One account ended at
- -32
- dollars. Another ended at
- -47
- dollars. Which balance is greater?

Hint Which is closer to 0?

Show your work.

Greater _____

- 4.
- Freezer Check.**
- One freezer holds at
- -18
- degrees. Another holds at
- 5
- degrees. Which reading is colder?

Show your work.

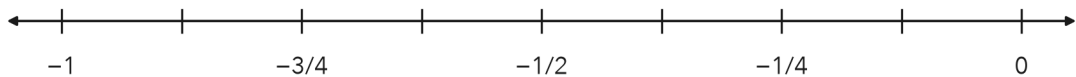
Colder _____

- 5.
- Two Cave Floors.**
- Cave A has a floor at
- -24
- m. Cave B has a floor at
- -9
- m. Write a comparison using
- $<$
- or
- $>$
- . Which cave is deeper?

Hint The deeper floor is farther left.

Deeper _____

6. **Eighths.** Plot $-7/8$ and $-1/8$. Which number is greater?



Greater _____

7. **Amina’s Claim.** Amina says $-0.9 > -0.09$ because nine tenths is more than nine hundredths. Is she right? Which number is greater?

Hint Write -0.9 as -0.90 .

Show your work.

Greater _____

8. **Three Accounts.** Three accounts hold -45 , 12 , and -8 dollars.

- a. Order the balances from least to greatest. _____
- b. Which balance has the largest absolute value? _____
- c. Why is that balance not the greatest? _____

Show your work.

FIND THE ERROR explain the mistake, then fix it

9. **Camila’s Work.** Camila wrote $-15 > -4$. She said 15 comes after 4 when you count, so -15 must be greater. Explain her mistake. Then write the correct comparison.

TEST PRACTICE choose the best answer

10. Which symbol makes -6 ____ -2 true?
A $<$ **B** $>$ **C** $=$ **D** none of these
11. On a number line, the greater number is always –
A farther left **B** farther right **C** farther from 0 **D** negative

SELF-CHECK CODE find your answer · copy its letter into that problem’s box · spell a math word

$-6 > -11$ C	-0.9 X	-0.09 R	-47 Q	-32 O	$-1/8$ A	-45 E	$-7/8$ Z	-18 M	Cave A P
#2b		#3	#4	#5	#6	#7	#8b		

CHECK YOURSELF ☐ I can do this on my own ☐ I need a little help ☐ I have a question about # _____